Ranking models for UK All-Weather Flat handicaps: the exploded conditional logit and the place market

Paper 2b in a series on UK All-Weather racing outcomes

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Paper 2a established a win-prediction model on UK All-Weather Flat handicaps using an extended feature set and race-level interactions. This paper holds that feature set fixed and asks three questions in turn. First (Q1), refitted under a ranking objective — the exploded conditional logit, equivalent to maximising the Plackett–Luce likelihood over the top-3 finishing order — how well does the model predict that order? We measure it with P1_rank and a top-3 Brier score, taking a discounted-Harville place-market baseline derived from starting prices as the yardstick: the model predicts the finishing order with clear skill above chance, and the market scores higher on both metrics. Second (Q2), because the model is now fitted on richer depth-3 ranking supervision, how does it pick *winners* compared with 2a's win-fitted model? On the depth-1 win backtest it returns -17.4% against 2a's -25.4%; a paired race-level bootstrap puts the difference at +8.0 pp (90% CI +1.0 to +14.8 pp), an interval that lies above zero but close to it; both models remain a net loss against the market. Third (Q3), turned into place and each-way bets on the real industry-SP book — the same margin-laden prices the win backtest pays out against — the model's place selections return -17.6%, alongside the win backtest's -17.4% and below a bet-everything baseline of -14.1%; the model's selections return less than indiscriminate betting, consistent with the ranking result.

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1 Introduction

Paper 2a fixed the prediction objective as the *win* and asked how far an enriched feature set could narrow the gap to the betting market on UK All-Weather Flat handicaps. This paper holds that feature set fixed and asks three questions in order.

Q1 — ranking performance. Refitting the same features from scratch under a ranking objective — the exploded conditional logit, which maximises the Plackett–Luce likelihood over the top-3 finishing order — how well does the model predict that order? We re-estimate the whole model on the ranking objective, noting where its feature picture departs from paper 2a's (the draw×course block, re-searched here, retains three courses where 2a kept two; Section 3), and measure performance with two ranking metrics, P1_rank and a top-3 Brier score, against a discounted-Harville place-market baseline. The model predicts the finishing order with clear skill above chance, and the market scores higher on both metrics.

Q2 — win performance as a byproduct. Because the model is now fitted on richer depth-3 ranking supervision rather than depth-1 win supervision, how does it pick *winners* compared with 2a's win-fitted model? On the depth-1 win backtest the ranking-fitted model returns a higher (less negative) ROI, by a margin a paired bootstrap places above zero though close to it; both models remain a net loss against the market.

Q3 — betting value. Does the model's place-probability accuracy translate into betting value? Turned into place and each-way bets priced on the real industry-SP book — the same margin-laden basis as the win backtest — do the model's selections return more than betting indiscriminately? They return less, and on the place bet the return sits alongside the win backtest's, consistent with the ranking result.

Section 2 points to paper 2a for the shared data and feature set and constructs the place-market baseline; Section 3 fits the ranking model and re-searches the course×draw block; Section 4 answers Q1, then Q2, then Q3; and Section 5 draws the threads together.

2 Data and scope

The data, the race-selection rules, the train/test split and the feature set are identical to paper 2a; we refer the reader there (paper 2a, §2) rather than repeat them. The one genuinely new ingredient this paper needs is a *place-market* baseline, constructed below.

2.1 Market baseline: discounted Harville

The win-market baseline used in papers 1 and 2a — starting-price implied win probability, over-round adjusted — cannot be directly applied to a top-3 ranking objective. We require market-implied *place* probabilities for each horse.

We derive these from win starting prices using the discounted Harville formula (Lo and Bacon-Shone 1994, 2008). Pure Harville (Harville 1973) rests on a single assumption: once the winner is removed, the remaining horses "re-race" with win probabilities proportional to their original ones — the field's relative strengths are unchanged by the outcome. Under that assumption the probability that horse j finishes second given horse i won is $p_j/(1-p_i)$, where p_k denotes win probability; the probability of a specific top-2 order (i, j) is $p_i\, p_j/(1-p_i)$, and the same conditioning extends to deeper rankings.

We use discounted Harville as the market baseline for both metrics — the P1_rank order probability and the Brier_place place probability. The reason to discount rather than use pure Harville is a known bias: pure Harville overstates the place chances of favourites and understates those of longer-priced horses, the favourite–longshot bias operating in the place dimension. That bias sits in the marginal place probabilities and carries through into the joint top-3 order probabilities, so pure Harville would be a biased baseline for both metrics. The discounted form of Lo & Bacon-Shone corrects it by replacing each win probability p_k with p_k^α $(\alpha < 1)$ inside the conditional steps before renormalising, with a smaller α at deeper positions; we use $\alpha = 0.80$ for second place and $\alpha = 0.65$ for third, Lo & Bacon-Shone's recommended values for starting-price inputs.

3 The exploded conditional logit and the course×draw search

The win model of paper 2a uses only one bit of each race's result: who won. The finishing order carries more. The **exploded conditional logit** (Henery 1981) uses it by treating the observed order as a sequence of conditional win problems — the winner is chosen from the full field, the runner-up from the field minus the winner, the third from the field minus the first two — and pooling those sub-races into a single conditional-logit fit, which is exactly maximising the Plackett–Luce likelihood. We truncate at depth $k = 3$: only the top-3 order is modelled, the tail of the field being largely noise (Henery 1981). The algebra and the Plackett–Luce equivalence are derived in Section 6.

The base specification is **identical to paper 2a's reduced win model** — the same 17 terms, the same zero-plus-slope position encoding, the same imputed `or_relative`/`or_missing` — so any difference at this stage is the explosion, not the features. The draw×course block is then searched **independently** of paper 2a (below); we do not assume 2a's surviving courses carry over.

3.1 From win to ranking: the exploded construction

For each race we form $k = 3$ sub-races: choice set s is the horse that finished sth together with every horse not yet ranked — all finishers placed s-or-worse plus any non-finishers, which never get ranked and so stay in the pool at every depth. The depth-1 sub-race is therefore exactly the win model's choice set (the full field, winner = 1st), which makes the two models comparable at depth 1; each sub-race has exactly one winner.

Two race filters apply. Races with any missing value in the modelling columns are dropped (the same rule as the win model), and races whose top-3 positions are not each recorded exactly once — a dead-heat for 2nd or 3rd, or a gap — cannot be unambiguously exploded and are dropped too. Finishing position is the official placing, taking the amended placing where a result was later revised.

Table 1: Exploded training data (carrying the draw-course columns): 5,022 valid races become 15,066 choice sets (3 per race) over 122,844 runner-rows.

Sub-race	Runner-rows
depth 1 (full field)	45970
depth 2 (field − 1st)	40948
depth 3 (field − 1st,2nd)	35926

The runner-row count falls by one per race at each successive depth: the same races contribute to all three sub-races, but each deeper sub-race drops the horses already placed above it (depth

2 removes the winner, depth 3 the first two), so the choice set shrinks by one runner each time.

3.2 Fitting the ranking model

We carry paper 2a's reduced 17-feature specification as the base — the feature-by-feature selection was settled there — and re-estimate it under the ranking objective, entering the four-course draw×course block (Kempton, Lingfield, Southwell, Wolverhampton) from the start. The draw block is then reduced by the same per-term Wald $p < 0.05$ rule paper 2a applies to its win model, run fresh on this fit: the least significant non-significant course slope is dropped, the model refit, and the step repeated until every surviving slope clears the threshold.

Re-estimating the base under the explosion moves the coefficients in two ways, shown against paper 2a's win fit on the identical specification.

Table 2: Paper 2a reduced win model (depth-1) vs the exploded base fit (depth-3 Plackett–Luce), same 17-term specification. Estimates and Wald z-values side by side.

Block	Term	Win est.	Win z	Exploded est.	Exploded z
1. Position (zero-plus-slope)	pos_lag1_zero	-0.8239	-15.69	-0.7361	-23.59
1. Position (zero-plus-slope)	pos_lag1_nonzero	-0.1494	-8.19	-0.1179	-10.99
1. Position (zero-plus-slope)	pos_lag2_zero	-0.3258	-6.02	-0.3148	-9.86
1. Position (zero-plus-slope)	pos_lag2_nonzero	-0.0495	-2.61	-0.0335	-3.02
2. Form / history	daysLTO (log)	-0.1277	-7.13	-0.1525	-14.71
2. Form / history	trainerSR	3.3172	7.93	3.0582	12.26
2. Form / history	sireSR	1.8271	4.33	1.0975	4.29
3. Horse characteristics	age_diff	-0.0915	-6.54	-0.0579	-7.36
3. Horse characteristics	entire	0.4411	7.69	0.3374	9.85
3. Horse characteristics	gelding	0.2433	5.37	0.2205	8.47
3. Horse characteristics	cheekpieces	-0.2690	-4.38	-0.1406	-4.16
4. New features	jockeySR	4.0923	8.48	2.7528	9.72
4. New features	rel_weight	-0.0309	-6.60	-0.0095	-3.55
4. New features	or_relative	0.0783	18.36	0.0455	19.52
4. New features	or_missing	-0.3845	-4.46	-0.4646	-9.27
4. New features	trainer_aw_premium	1.7298	3.91	1.3906	5.33
4. New features	has_wins	0.1209	2.16	0.0939	2.93

The estimates shrink — most coefficients are smaller in magnitude under the explosion, because the win model loads each feature with all of its power to separate the *winner* from the field, whereas the exploded model must make the same coefficient explain the 1-vs-rest, 2-vs-rest and 3-vs-rest contrasts at once. The Wald z-values mostly move the other way, rising, as depth-3 explosion roughly triples the number of choice problems the same races supply; the form and position terms, which order the whole field rather than just crown the winner, change the most.

Table 3: Per-term reduction of the exploded draw×course block. Each step drops the least significant non-significant course slope and refits; the 1-df likelihood-ratio drop-test is reported alongside.

Step	Dropped	Wald p	From → to	LR stat	df	LR p
1	lingfield	0.91	4-course → 3-course	0.013	1	0.91

Only **Lingfield** drops (Wald $p = 0.91$; the 1-df LR drop-test agrees, $p = 0.91$). The surviving block is **Kempton, Southwell and Wolverhampton** — three courses, where paper 2a's win model retained only **Kempton and Southwell**. The difference is Wolverhampton. On the exploded fit Wolverhampton's draw slope is -0.182 ($p = 0.0003$); in paper 2a's win model it was -0.160 ($p \approx 0.066$) — the **same sign and similar magnitude, estimated far more sharply**. The reading is the narrow one: the ranking objective estimates the same draw effect paper 2a saw, sharply enough to retain it, rather than uncovering a new effect the win model missed. The negative slope (an advantage to lower draws) is the direction 2a already found and is directionally consistent with a tight, left-handed All-Weather track; it is a single distance-pooled slope, so it is an average direction across trips, not a trip-specific law.

The final paper-2b model is the exploded base plus the three retained course-draw slopes — 20 coefficients in all.

Table 4: Draw×course coefficients in the full four-course block (the selection-stage exploded fit). Kempton, Southwell and Wolverhampton are individually significant and retained; Lingfield is not and is dropped, leaving the three-course final model.

Course	Estimate	Std. error	z	p-value	Retained
Kempton	0.1368	0.0510	2.68	0.007280	yes
Lingfield	-0.0061	0.0541	-0.11	0.910000	dropped
Southwell	0.1683	0.0698	2.41	0.015900	yes
Wolverhampton	-0.1816	0.0505	-3.60	0.000323	yes

4 Evaluation on ranking metrics

This paper asks three questions, in order. **Q1** is the ranking question the paper is built around: how well does the exploded model predict the top-3 finishing order, measured against the market? **Q2** is a byproduct: because the model is fitted on richer (depth-3) ranking supervision rather than depth-1 win supervision, how does it pick *winners* compared with paper 2a's win-fitted model? **Q3** is the applied question: turned into bets, does that place-probability accuracy translate into betting value? The three are answered with different baselines and reported separately.

4.1 Ranking metrics

We evaluate ranking on two metrics, both computed out-of-sample on the held-out test races.

P1_rank (ranking discrimination): the geometric mean, over test races, of the Plackett–Luce probability assigned to the observed top-3 finishing order. Computed as the exponentiated mean log-likelihood per race to avoid numerical underflow:

$$\mathrm{P1}_{\mathrm{rank}} = \exp\left(\frac{1}{n}\sum_{i} \log P(j_1^i, j_2^i, j_3^i)\right),$$

where $P(j_1, j_2, j_3)$ is the depth-3 Plackett–Luce probability of the observed order under the model or market. Higher is better. This is the ranking analogue of Owen's P1.

Brier_place (place calibration): the mean squared error between the predicted place probability and the binary place outcome (top-3 or not), averaged over all runner-race observations in the test set:

$$\mathrm{Brier}_{\mathrm{place}} = \frac{1}{N}\sum_{i,j} \left(\mathrm{place}_{ij} - p_{ij}^{\mathrm{place}}\right)^2,$$

where $\mathrm{place}_{ij} = 1$ if horse j finished in the top 3 in race i, and p_{ij}^{place} is the model- or market-implied probability of a top-3 finish. Lower is better; a proper scoring rule, analogous to Owen's P2.

Both market baselines use discounted Harville ($\alpha = 0.80$ for the second-place conditional, 0.65 for the third; Section 2): the place probabilities for Brier_place and the depth-3 order probability for P1_rank are both built from the same discounted starting-price win probabilities. The model side of each metric uses its own ($\alpha = 1$) Plackett–Luce implication.

4.2 Q1 — Ranking performance against the market

The model is scored on 2,183 test races[1] against the discounted-Harville market baseline.

Table 5: Ranking metrics on the 2,183 scorable test races: exploded model vs discounted-Harville market baseline.

Metric	Model	Market	Better	Winner
P1_rank	0.00402	0.005427	higher	market
Brier_place	0.20130	0.187500	lower	market

The model predicts the top-3 order with clear skill above chance. A random-guessing baseline scores about 0.003 on P1_rank — roughly the reciprocal of the number of possible top-3 orderings in a typical field — and about 0.23 on Brier_place, the score from predicting every horse's place chance at the base rate. The model scores 0.00402 on P1_rank (higher is better) and 0.201 on Brier_place (lower is better), well inside the skilled range on both. Measured against the discounted-Harville market baseline, the market scores higher on each — 0.00543 on P1_rank and 0.188 on Brier_place. The model has genuine ranking skill; the market, here as in papers 1 and 2a, sets the higher mark, consistent with the starting price aggregating information the pre-race feature set does not contain.

4.3 Q2 — Win performance as a byproduct

The same fully-specified three-course exploded model can be judged on the *win* objective, by reading off its depth-1 win probabilities and running paper 2a's exact betting backtest. The question is not whether it beats the market — neither model does — but whether ranking supervision makes it a better or worse winner-picker than paper 2a's win-fitted model.

Table 6: Exploded (ranking-fitted) model on the depth-1 win backtest, AW test races at Owen's exact thresholds.

Prob threshold	Ratio threshold	Bets	Wins	Gross return	Profit	ROI
0.15	1.3	1426	160	1177.50	-248.50	-17.4%

The ranking-fitted model returns a naive win ROI of **-17.4%**, against paper 2a's win-fitted headline of **-25.4%** — both struck on the identical 2,193-race test set. Because the two models are evaluated on the same races we can pair the resampling: a race-level ROI-difference

[1] Q1 scores 2,183 races and Q2 below scores 2,193; the difference is that the ranking metrics require a clean, unambiguous top-3 (positions 1–3 each recorded exactly once) and a complete market win-probability vector, a filter the depth-1 win backtest in Q2 does not need.

bootstrap (B = 2000, seed 42) restricted to the common races gives a point difference of **+8.0 pp** in the ranking model's favour, with a 90% interval of [**+1.0 pp, +14.8 pp**].

Table 7: Paired race-level ROI-difference bootstrap (2b ranking-fitted minus 2a win-fitted), restricted to the common test races.

Contrast	ROI diff.	90% CI	Common races
2b ranking-fitted - 2a win-fitted	+8.0 pp	[+1.0 pp, +14.8 pp]	2,193

The 90% interval lies above zero, so on these test races the ranking-fitted model's win ROI is distinguishable from the win-fitted model's. The reading is that depth-3 ranking supervision yields a somewhat better winner-picker than depth-1 win supervision. Two things keep it in proportion: the lower bound of the interval (+1.0 pp) sits close to zero, so this is a distinguishable difference rather than a wide one; and both models return a negative ROI (-17.4% here), so the gain is relative to paper 2a, not a profit against the market. The ratio-threshold sweep below shows the win ROI across selection thresholds.

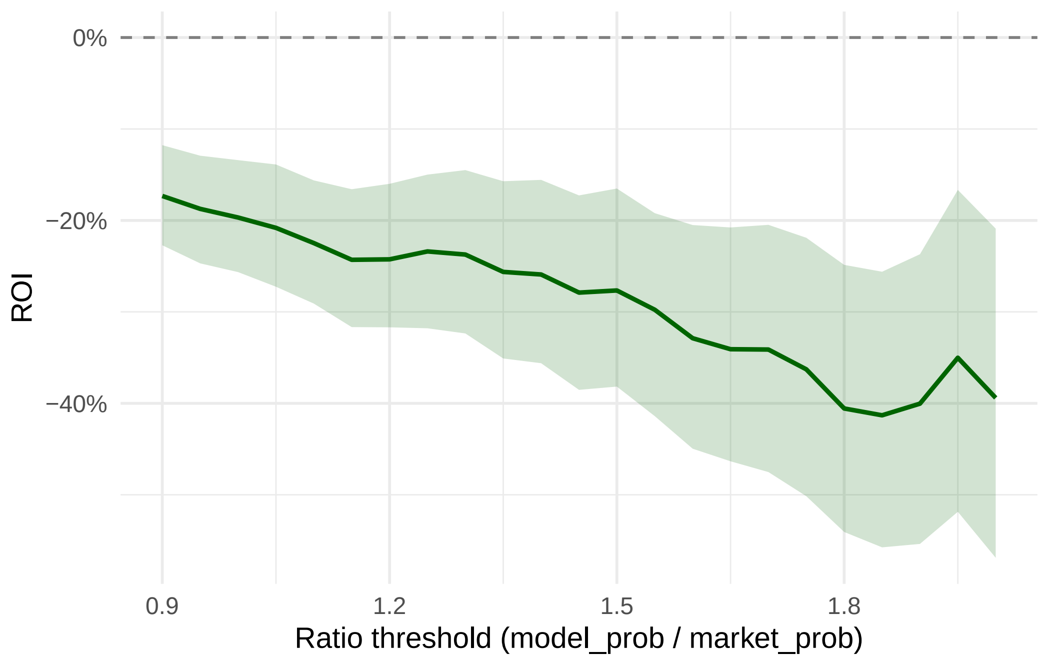

Figure 1: Depth-1 win ROI vs ratio threshold on the AW test set (ranking-fitted exploded model), 90% bootstrap CI shaded. Model-probability filter held at 0.13; dashed line at zero marks break-even.

4.4 Q3 — Does place-probability accuracy translate into betting value?

Q1 and Q2 read the model against the market on scoring rules. The applied question is whether that accuracy translates into betting value: priced and selected as bets, do the model's probabilities return more than the market's? We ask it of two bet types — a horse to finish in the top three (**place**) and the same horse staked win-and-place at standard each-way terms (**each-way**) — each priced and selected entirely from win starting prices, with no separate place or each-way market prices used or needed.

Selection and payout play separate roles. **Selection** — the value ratio that decides which bets to place — uses the discounted Harville probabilities ($\alpha = 0.80/0.65$), the bias-corrected place construction Q1 is scored against; a bet is placed when the discounted model probability exceeds the discounted market probability by a swept ratio threshold τ, Owen's naive rule from papers 1 and 2a applied to these bets. **Payout** — what a winning bet returns — is the real industry-SP book, the same margin-laden starting prices the Q2 win backtest pays out against, so Q3 sits on one footing with Q2. The place price is derived from the raw SP-implied win probabilities via Harville, inheriting the win book's margin; the each-way win leg pays the raw SP win odds and the place leg one fifth of those odds on a top-3 finish, standard terms. The industry SP book on these races carries an observed average over-round of 16% — measured from the data, not assumed — which every payout reflects.

Because the payout carries the real margin, the neutral reference — betting indiscriminately at these prices — is itself negative, not zero, and each model ROI is read against that bet-all baseline. This is exactly the basis of the Q2 win backtest: the headline place return below sits alongside Q2's -17.4%. Alongside it we retain a zero-margin *fair-book* figure — the same bets priced at a normalised pure-Harville book — as a secondary reference that isolates the model's selection skill from the margin.

Table 8: Place and each-way value bets at Owen's naive rule (ratio > 1.3). Headline columns price the payout at the real industry-SP book (observed average over-round 16%), the same basis as the Q2 win backtest, so the bet-all baseline is negative; the fair-book columns (zero-margin pure Harville) are a secondary reference isolating model skill from the margin. Selection is identical across both bases.

Bet type	Races	Bets	Model ROI (SP)	Bet-all (SP)	Model − baseline	Model ROI (fair)	Bet-all (fair)
Place	2,188	5,183	-17.6%	-14.1%	-3.5 pp	-3.6%	0.3%
Each-way	2,188	6,672	-27.6%	-18.5%	-9.1 pp	-16.8%	-7.3%

Place. On the real-SP book, betting every runner to place returns -14.1% — the over-round, carried through into the derived place price. The model's place value-selections return -17.6% on 5,183 bets, about 3.5 percentage points below that baseline, and the headline return sits alongside the Q2 win backtest's -17.4%: the market's place prices are the more accurate, the same finding Q1's Brier_place reported. Three things make the reading robust. Place payouts are short, so no single result drives it — dropping the largest winning bet leaves the return at -18.1% — and the bootstrap intervals are tight. The model–baseline gap is insensitive to the margin: scaling the SP over-round across a ±5-point range (11% to 21%) holds it at about -3.5 pp (-3.3 pp to -3.6 pp), so the result does not depend on the exact margin. And the sweep (Figure 2) shows the model's return falling as the ratio threshold rises, so the place bets it most strongly prefers to the market are the ones it does worst on.

Each-way. Betting every runner each-way returns -18.5%, a baseline pushed below zero by two effects together: the real SP margin, as for place, and the structural each-way terms, the place leg paying one fifth of the win odds — shorter than the fair place price. The model returns -27.6%, about 9.1 percentage points below that baseline; as with place, its selections sit below indiscriminate betting.

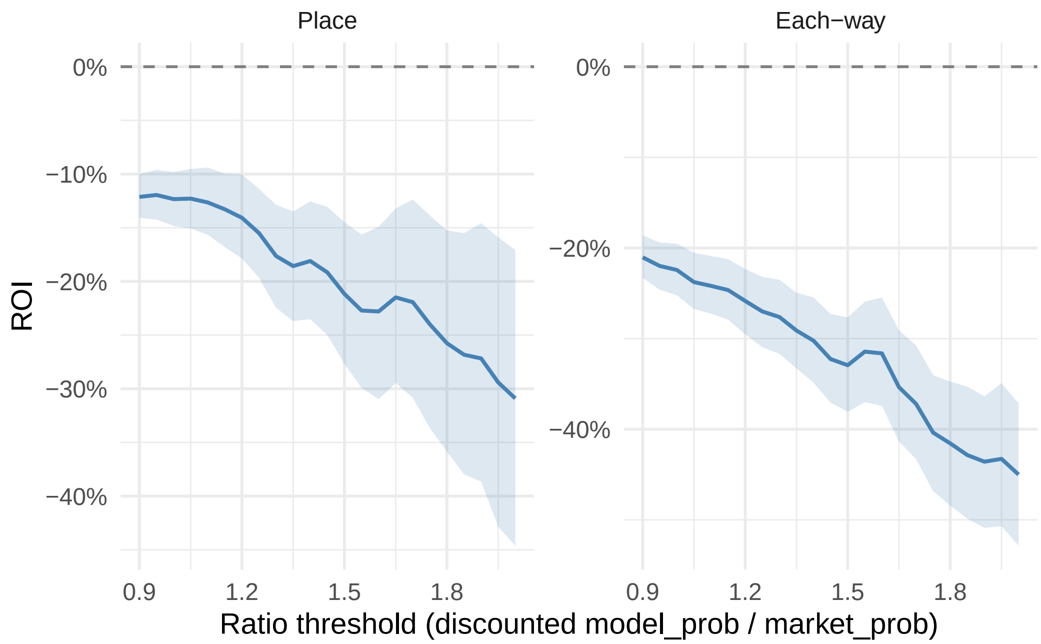

Figure 2: Real-SP ROI vs ratio threshold for the place and each-way value bets, 90% bootstrap CI shaded; dashed line at break-even. Free vertical scales. Both fall further below the (negative) real-SP baseline as selection tightens.

5 Discussion

The exploded model predicts the top-3 finishing order with skill clearly above chance, yet on neither ranking metric does it match the market. That is the same pattern papers 1 and 2a found on the win objective — a model with genuine signal that the starting price nonetheless out-scores — now repeated on the ranking objective. The reading carries over unchanged: the market price aggregates information the pre-race feature set does not contain, and moving from a win objective to a ranking one does not narrow that gap. Turned into bets, the same picture holds: priced on the real industry-SP book, the model's place selections return below both a zero-margin fair book and indiscriminate betting at the market's own prices, so its disagreements with the market are, where they can be priced, on the losing side.

The second question applies the ranking-fitted model to the win objective, and here the result is new to the series. Trained on the finishing order rather than on the winner alone, the model is a better winner-picker than paper 2a's model: a paired race-level bootstrap puts the gain at +8.0 pp, with a 90% interval that lies above zero — the first distinguishable improvement across the three papers. The mechanism is supervision, not specification. Depth-3 explosion turns each race into three nested choice problems — the winner from the full field, the runner-up from the rest, the third from what remains — so the same races and the same functional form are estimated from roughly three times as many contrasts, and the coefficients come back sharper. Wolverhampton is the clean illustration: its draw term was too noisy to retain under win supervision (Wald $p \approx 0.066$) but is estimated sharply under the ranking objective ($p = 0.0003$), with the same sign and a similar magnitude. The reading is the narrow one — the ranking objective estimates the same draw effect paper 2a saw, sharply enough to retain it — not a new effect uncovered. None of this makes the model profitable, and the gap to the market is no smaller; the improvement is over paper 2a, not over the price.

Paper 3 changes the model class. A tree-based learner accommodates non-linearities and feature interactions directly — among them a horse's going affinity, which the conditional logit can express only through explicit interaction terms — without the modeller specifying each one in advance. Evaluating the exotic markets — the exacta and trifecta — is left to future work: their real prices are the CSF and Tote-pool dividends, which this paper does not have.

6 Appendix: Derivations

This appendix derives the Plackett–Luce model and its exploded-logit representation — the ranking machinery fitted in Section 3 — and the Harville place-probability formula behind the market baseline of Section 2. It builds on the conditional logit derived in paper 1's appendix, whose notation we reuse: race i is the chooser, horse j the alternative, the linear predictor for horse j in race i is $z_{ij} = \sum_l \beta_l x_{ilj}$ over alternative-specific features x_{ilj}, and win probabilities are given by the softmax $p_{ij} = e^{z_{ij}} / \sum_m e^{z_{im}}$.

6.1 From conditional logit to Plackett–Luce

The conditional-logit model estimates only the probability that a particular horse *wins* a race. In practice, we observe a full or partial *ranking* of the field, which contains more information than the win alone. The Plackett–Luce model (Plackett 1975) extends conditional logit to rankings by treating the observed order of finishers as a sequence of conditional win probabilities: horse j_1 wins the race, then horse j_2 wins among the remainder, and so on.

Formally, let $j_1, j_2, \ldots, j_k$ denote the top k finishers in race i, in order. The Plackett–Luce probability of this partial ranking is

$$P(j_1, j_2, \ldots, j_k) = \prod_{s=1}^{k} \frac{e^{z_{ij_s}}}{\sum_{r=s}^{J} e^{z_{ij_r}}},$$

where the denominator at step s sums over all horses not yet ranked. Setting $k = J$ gives the probability of a full ranking; setting $k < J$ gives the truncated version, in which only the top k finishers are modelled. In racing, truncation is natural: while the top few finishers are separated by genuine differences in ability and conditions on the day, the ordering among the tail of the field is largely noise and carries little useful signal (Henery 1981). Truncating at k discards this noise; paper 2b takes $k = 3$.

The log-likelihood over a set of races is

$$\sum_{i=1}^{n} \sum_{s=1}^{k} \left(z_{ij_s} - \log \sum_{r=s}^{J} e^{z_{ij_r}} \right).$$

As in the conditional-logit case, this is concave in the coefficients, so any local maximum is the global maximum.

6.1.1 Exploded conditional logit and Plackett–Luce equivalence

The Plackett–Luce likelihood has a natural interpretation in terms of conditional logit. Each factor in the product

$$\prod_{s=1}^{k} \frac{e^{z_{ij_s}}}{\sum_{r=s}^{J} e^{z_{ij_r}}}$$

is precisely the conditional-logit probability that horse j_s wins *among the horses remaining at position s*. So the Plackett–Luce likelihood is equivalent to the likelihood of k separate conditional-logit problems, one for each position in the ranking, each defined on a shrinking choice set. This is known as the *exploded logit*: each race is exploded into k conditional-logit observations.

Concretely, define $\mathcal{C}_{is} = \{j_s, j_{s+1}, \ldots, j_J\}$ as the choice set at position s — the horses not yet ranked. Then the Plackett–Luce log-likelihood can be written as

$$\sum_{i=1}^{n} \sum_{s=1}^{k} \left(z_{ij_s} - \log \sum_{j \in \mathcal{C}_{is}} e^{z_{ij}} \right),$$

which is identical in form to the conditional-logit log-likelihood (McFadden 1974), summed over nk observations.

In practice, each race is exploded into k fictitious races: the first is the real race with the full field; the second is a fictitious race consisting only of the horses that finished second or worse; the third a fictitious race consisting only of those that finished third or worse; and so on. These nk races — real and fictitious — are pooled into a single dataset, and a single conditional-logit model is fit across the whole pool. By making fuller use of the finishing order rather than just the win, this yields more precise estimates of β. The construction is implemented in `prepare_exploded_data()`, and the fitted result is Section 3's exploded ranking model.

6.2 Harville order and place probabilities: the market baseline

The exploded logit above is the *model* side of Section 4; its *market* side needs place probabilities — the chance a horse finishes in the top k — from win starting prices alone. Harville (Harville 1973) supplies them under a single assumption: once the winner is removed, the remaining horses re-race with win probabilities proportional to their original ones, so the field's relative strengths are unchanged by the result.

Order probability. Let $p_1, \ldots, p_J$ be the win probabilities. Under the assumption, given that horse i won, horse j's conditional probability of winning the "race for second" is $p_j/(1 - p_i)$ — its share of the strength remaining once i is removed. Iterating down the order, the probability of a full finishing sequence $(j_1, j_2, \ldots, j_J)$ is

$$P(j_1, j_2, \ldots, j_J) = \prod_{s=1}^{J} \frac{p_{j_s}}{1 - \sum_{r<s} p_{j_r}},$$

each factor being the leader's share of the strength not yet placed.

Equivalence with Plackett–Luce. This is exactly the Plackett–Luce order probability derived above. Writing $p_j = e^{z_j} / \sum_m e^{z_m}$, the denominator $1 - \sum_{r<s} p_{j_r}$ equals $\left(\sum_{r \geq s} e^{z_{j_r}}\right) / \sum_m e^{z_m}$; the normaliser $\sum_m e^{z_m}$ cancels against the one in the numerator's p_{j_s}, leaving $\prod_s e^{z_{j_s}} / \sum_{r \geq s} e^{z_{j_r}}$ — the Plackett–Luce expression. Undiscounted Harville and Plackett–Luce are the same model, reached from different starting points (sequential conditioning on win probabilities versus a ranking likelihood). This is why an undiscounted Harville baseline cannot judge a Plackett–Luce-fitted model: it *is* that model's formula, evaluated at the market's win probabilities rather than the model's.

Place probability. The marginal probability that a given horse finishes in the top k is the sum of the order probability over every sequence that places it among the first k. Harville's product form keeps this tractable: for $k = 3$ the top-3 marginal for horse j is

$$P(j \in \text{top } 3) = p_j + \sum_{i \neq j} p_i \frac{p_j}{1 - p_i} + \sum_{i \neq j} \sum_{m \neq i,j} p_i \frac{p_m}{1 - p_i} \frac{p_j}{1 - p_i - p_m},$$

the three terms being the probabilities that j finishes first, second and third. The inner sums have only $O(J)$ and $O(J^2)$ terms, so the place probabilities are computed in closed form without enumerating the $J!$ orders.

Discounted Harville. Pure Harville is known to overstate favourites' place chances and understate longshots' — the favourite–longshot bias in the place dimension — because the sequential-conditional assumption hands too much of the remaining probability mass to the strongest horse at each step. Lo & Bacon-Shone (Lo and Bacon-Shone 1994, 2008) correct it by replacing p_i with p_i^α ($\alpha < 1$, and smaller at deeper positions) inside the conditional steps before renormalising — $\alpha = 0.80$ for second place and 0.65 for third in this paper. Because each conditional is then built from the discounted strengths p_i^α rather than the win probabilities themselves, the cancellation that produced the Plackett–Luce form no longer happens: the discounted order probability is not a product of plain win probabilities, and the discounted Harville and Plackett–Luce models cease to coincide. That break is what makes the discounted baseline a genuine yardstick for the exploded model rather than a re-expression of it.

The mixed discrete-choice derivation, used in paper 2a for race-level interactions, is given in paper 2a's appendix.

7 Appendix: Software and reproducibility

The software stack — R, the tidyverse, {mlogit} for the conditional-logit and exploded fits, {DBI}/{RMariaDB} for the Smartform database, {renv} for package pinning, and Quarto for typesetting — is the same as paper 1's, whose appendix B describes it in full, and we do not repeat that here. As there, the whole analysis is a {targets} pipeline: each step in the data → features → model → render chain is a `tar_target()`, intermediate results are cached on disk, and a single `targets::tar_make()` from the project root rebuilds every stale step and re-renders the paper. Fitting is on the training split only; every out-of-sample figure is computed on the held-out test races.

Acknowledgments

I have been interested in modelling horse racing since first acquiring the Smartform database more than a decade ago. The project has been picked up and put down many times, the obstacle almost always being the gap between an idea and a running implementation. What is different

now is that working alongside Claude Code (Anthropic's coding agent), with web-based Claude as a research companion, has compressed that gap dramatically. I am grateful to my brother, Michael Gillen, for the suggestion to build the model on All-Weather races — the foundation of the scope choices set out in Section 2. The design choices — dataset, scope, model, comparison, editorial line — are mine. The implementation, the bulk of the draft prose, and a great deal of the verification were Claude Code's, under my direction; every output was reviewed and edited before it reached the paper. I do not yet know what general lessons this experiment points to about the value of human–AI collaboration in applied statistics. That uncertainty is part of why I am doing the work, and these papers are partly a record of that experiment.

References

Harville, David A. 1973. "Assigning Probabilities to the Outcomes of Multi-Entry Competitions." *Journal of the American Statistical Association* 68 (342): 312–16.

Henery, R. J. 1981. "Permutation Probabilities as Models for Horse Races." *Journal of the Royal Statistical Society: Series B (Methodological)* 43 (1): 86–91.

Lo, Victor S. Y., and John Bacon-Shone. 1994. "A Comparison Between Two Models for Predicting Ordering Probabilities in Multiple-Entry Competitions." *Journal of the Royal Statistical Society: Series D (The Statistician)* 43 (2): 317–27.

Lo, Victor S. Y., and John Bacon-Shone. 2008. "Approximating the Ordering Probabilities of Multi-Entry Competitions by a Simple Method." In *Handbook of Sports and Lottery Markets*, edited by Donald B. Hausch and William T. Ziemba. North-Holland (Elsevier).

McFadden, Daniel. 1974. *Conditional Logit Analysis of Qualitative Choice Behavior*. (New York), 105–42.

Plackett, R. L. 1975. "The Analysis of Permutations." *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 24 (2): 193–202.